

The Same Word Twice

Team Members: _____

Today you learn two things that share a name. A **conditional statement** in Python is an `if`: do this when a condition holds. A **conditional probability** is a chance computed when a condition holds. That is not a coincidence, and by the end of class you should be able to say exactly what the two have in common.

Part 1. Predicate Golf

Here are twelve patients. Four of them are marked.

patient	age	sex	cp	thalach
1	62	M	0	165
2	41	F	2	172
3	58	M	1	143
4	45	M	0	148
5	67	F	3	132
6	52	F	2	168
7	71	M	0	125
8	49	M	1	155
9	54	F	0	139
10	60	M	2	158
11	44	F	3	176
12	56	M	1	151

Write a boolean expression that selects the four marked patients — all of them, and nobody else. Use column names, comparisons, and `&` (and), `|` (or), `~` (not). Parentheses around each comparison are required: `(age < 50) & (thalach > 160)`.

Scoring: fewest characters wins, spaces not counted. No single comparison can do it, so do not waste time looking for one.

1.1 Your expression:

Character count: _____ Winning count in the room: _____

1.2 Another team's expression selects the same four patients but looks nothing like yours — theirs is built from `~` and yours is not, or the other way round. Write theirs here and explain why both are correct.

1.3 In Python, `heart.column('age') > 55` actually is — one answer, or many?

Part 2. Thirty-Six Outcomes

Roll two dice, one at a time. The row is the first die, the column is the second, and the cell is the total. All thirty-six cells are equally likely.

	1	2	3	4	5	6
1	2	3	4	5	6	7
2	3	4	5	6	7	8
3	4	5	6	7	8	9
4	5	6	7	8	9	10
5	6	7	8	9	10	11
6	7	8	9	10	11	12

rows: first die

columns: second die

2.1 Circle every cell containing a 7.

$P(\text{sum is 7}) = \frac{\quad}{36} = \frac{\quad}{\quad}$

2.2 Now cover everything except one row — you have been told what the first die was. Count the 7s in each row.

first die 1: $\frac{\quad}{6}$ 2: $\frac{\quad}{6}$ 3: $\frac{\quad}{6}$ 4: $\frac{\quad}{6}$ 5: $\frac{\quad}{6}$ 6: $\frac{\quad}{6}$

Knowing the first die changes your answer by how much? This is what independence looks like on paper.

2.3 In craps you win immediately on a 7 or an 11. Put a box around the 11s as well.

$$P(\text{win}) = \text{_____} / 36 = \text{_____}$$

2.4 Cover everything but one row again.

$$P(\text{win} \mid \text{first die is 3}) = \text{_____}/6 \qquad P(\text{win} \mid \text{first die is 5}) = \text{_____}/6$$

Those are not equal, and neither equals your answer to 2.3. Something changed between 2.2 and 2.4 — the dice did not. What did?

2.5 Now condition the other way. Of the winning cells, how many have a first die of 5?

$$P(\text{first die is 5} \mid \text{win}) = \text{_____}/\text{_____} = \text{_____} \qquad \text{compare with 2.4: } \text{_____}$$

Same cells on top. Different cells underneath. Say in one sentence what the denominator of a conditional probability is.

Part 3. Write a Diagnostic Rule

Open `Class09_Same_Word_Twice_Skeleton.ipynb`. The table holds 1,025 patients referred for cardiac testing; target is 1 if they turned out to have heart disease.

3.1 $P(\text{disease})$ for the whole table: _____ — an unconditional probability.

$$\text{3.2 } P(\text{disease} \mid \text{female}) = \text{_____} \qquad P(\text{female} \mid \text{disease}) = \text{_____}$$

Which of these is the grid from Part 2 again, and which cells changed?

Now the `if` statements. Your rule looks at a patient and returns `'flag'` — send them for further testing — or `'clear'`. Then you grade it two ways.

3.3 The mediocre `example_rule` in the notebook:

$$P(\text{flag} \mid \text{disease}) = \text{_____} \qquad P(\text{disease} \mid \text{flag}) = \text{_____}$$

3.4 The rule that flags everybody:

$P(\text{flag} \mid \text{disease}) = \underline{\hspace{2cm}}$ $P(\text{disease} \mid \text{flag}) = \underline{\hspace{2cm}}$

One of those is perfect. Explain in one sentence why that rule is nevertheless useless, and which of the two numbers caught it out.

3.5 Write your team's best rule here — the conditions, in words or in code — along with its two scores.

3.6 Whose rule is best? Before the room votes, your team has to say which of the two numbers it is competing on, and why.

We are competing on: $P(\text{flag} \mid \text{disease})$ $P(\text{disease} \mid \text{flag})$

Discussion

D1. An `if` statement in Python and the phrase “given that” in a probability are the same idea in two notations. Say what the idea is, using the word subset.

D2. A lab reports that its heart disease test is “95% accurate.” A patient gets a positive result and wants to know the chance they are sick. Which conditional probability is the lab quoting, which one does the patient want, and what else would you need to know to get from one to the other?

D3. You tuned your rule by testing it on the same 1,025 patients you used to grade it. What would you expect to happen to your two scores if we ran your rule on a thousand patients from a different hospital, and why?